Juwon Lee

Education R&D · Developer, Seoul Robot & AI Museum

Target roles: Education planning & development

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Summary

At the Seoul Robot & AI Museum, I plan education programs and build the software used in classes and exhibits, from student web apps to exhibit kiosks and staff tools. Three instructors I trained taught my smart-farm program for the Yangpyeong Education Office to a 15-student cohort at Gyeonggi Shared School in October 2026, and it ran for Geochang Daesung High School in November. My AI art exhibit generated 14,328 images on the museum floor from June to August 2026.

Experience

Education R&D · Developer · Seoul Robot & AI Museum (via operator FNJ Co., Ltd.) 2026.03 – Present

- Planned a 10-session, 20-hour AI and coding course for museum staff (one hour of theory and one hour of individual practice per session; five sessions are AI technology trend studies), running since November 4, 2026.
- Run the Picabot robot-arm drawing studio as a regular 30-minute hands-on AI session for visiting school groups; led its extended 50-minute class at the KT × Seoul City AI Future Camp (100 elementary students attended) in August 2026.
- Build internal tools behind exhibits and classes, including a staff scheduling app used daily by about 20 docents and part-time staff.

VQA Dataset Builder & QA Reviewer · Ubase, Inc. (dispatched to Select Star) 2025.11 – 2025.12

- Built and reviewed visual question-answering (VQA) datasets for LLM training on an ETRI (Electronics and Telecommunications Research Institute) project, including complex 3-hop reasoning Q&A sets.
- Sustained about 1.5× the team's average throughput while holding accuracy.

Youth Intern · Ministry of Science and ICT 2025.04 – 2025.08

- Supported improvements to the Pay-TV service quality evaluation and its satisfaction survey; drafted Broadcasting Act review materials.

Selected projects

Smart-Farm Education Program · Planning & development, solo 2026

IoT smart-farm and AI crop-diagnosis class commissioned by the Yangpyeong Education Office through the museum.

- Built the Next.js web app with server API routes on Vercel and Arduino (C++) firmware for the class's 15 ESP32 student boards (one per student), which write sensor readings straight to Supabase.

Mystery Playground RE:PLAY · Development, field tuning, operations tooling 2026

Ten-mission tablet game archiving the museum's 4th-floor special exhibition; five MediaPipe vision models run on the tablet.

- Traced misdrawn avatars to hats and headbands using the first 128 numbers-only visitor logs, then fixed the color, beard and curl rules and checked the change against a frozen eight-image benchmark.

Raimi's AI Art Lab · Planning, development, operations 2026

Guided prompt builder where visitors make AI artwork with the museum character Raimi.

- Replaced an off-the-shelf GPT service that froze mid-use with a custom web app and serverless backend; no freezes since the switch.

Exhibit Auto-Recovery · Diagnosis, build, rollout, handover 2026

Recovery system for a face- and voice-interaction robot exhibit (MaskBot) whose control software froze or exited mid-operation.

- Watchdog checks the program every 20 seconds and relaunches it within a minute of a crash or hang it can detect, with a two-stage check against false positives and a 5-minute pause after 3 restarts in a row.
- Recovery page with one button that any staff member can open on their own phone, reached over a Tailscale (WireGuard) mesh VPN with no port open to the internet.

Teaching & mentoring

2026 SW·AI Donghaeng Project (KOFAC) · Youth mentor · makerthon coaching, live online class mentoring (middle school) 2026.09 – 2026.10

KB LA School (middle school) · Middle-school mentor · twice a week, 80 min · outside work hours 2026.07 – Present

Geuruteogi Learning Mentoring · University mentor (6th cohort) · one-on-one study mentoring for a Grade-9 student · twice a week, 90 min · outside work hours 2026.02 – Present

Skills & languages

Planning & Education: Product Planning (PM) · Learning Design · Bilingual Exhibition Guiding (KO·EN)
Frontend & App: Next.js · React · TypeScript · JavaScript · Tailwind · PWA · Capacitor
Backend, Infra & Ops: Supabase · Firebase / Firestore · Flask · Serverless Functions · Cloudflare R2 · Vercel · Tailscale / WireGuard
AI & Computer Vision: Python · TensorFlow.js · MobileNet v2 · MediaPipe · VGG16 / CNN · OpenAI API
Hardware · IoT · Robotics: ESP32 · MicroPython · Arduino / C++ · Sensors
Languages: Korean (native) · English (fluent; TOEIC Speaking AL, OPIc AL)

Awards

Prize · 2025 GH Youth Build-Up Start-up Competition (GH Corp.) 2026.03

Excellence Award (2nd place, finals) · 2025 Science Museum & Community AI Hackathon (National Science Museum) 2025.09

Excellence Award · ICT·SW Women's Start-up Competition 2024.12

Education

Hankyong National University 2022.03 – 2027.02

Plant Life & Environment, double major in Software Convergence · expected to complete graduation requirements (graduation deferred)